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Luis R. Mercado-Diaz

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EDUCATION

University of Connecticut (UConn) — **Storrs, CT, USA**
Ph.D. in Biomedical Engineering (Specializing in Graph Signal Processing, Natural Language, Affective Computing and Wearable AI). (Anticipated) May. 2026

University of Connecticut — **Storrs, CT, USA**
M.S. in Computer Science (Information and Communication Technology). May. 2025

Instituto Tecnológico Metropolitano — **Medellin, Colombia**
B.S. in Biomedical Engineering (Graduated with The Highest Honors). Dec. 2021

EXPERTISE SPECIALIZED IN ARTIFICIAL INTELLIGENCE (AI)

- **Graph Neural Networks (GNN/GAT/GraphSAGE) [8+ years]**
- **Graph Signal Processing [8+ years]**
- **Knowledge Graph & Neo4J [8+ years]**
- **Python/PyTorch/TensorFlow [8+ years]**
- **Wearable Biosensing AI [8+ years]**
- **Multi-omics & Computational Biology [8+ years]**
- **Natural Language Processing (NLP) [7+ years]**
- **Large Language Model (LLM) [6+ years]**
- **Transformer Architectures (BERT/GPT) [6+ years]**
- **Deep Learning (CNN/LSTM/Attention) [7+ years]**
- **Cloud Deployment (Azure/Databricks) [6+ years]**
- **Real-time ML Pipeline Development [6+ years]**

NOTICIBLE EXPERIENCE WITH 10+ FIRST-AUTHORSHIP PUBLICATIONS

- Expertise in Graph Signal Processing, AI for Healthcare, and Biomedical Signal Analysis with 8+ years of research experience spanning industry and academia.
  - **Graph Neural Network Architecture Development**: Implemented and deployed Graph Attention Networks (GAT) and GraphSAGE models for multi-modal medical imaging analysis, achieving state-of-the-art stroke lesion detection with probabilistic attention mechanisms (Neurocomputing 2025). Developed novel GNN architectures for protein-protein interaction prediction and metabolic pathway analysis using GraphSAGE to compute interaction probabilities in biological networks.
  - **Knowledge Graph Pioneer for Biomedical Signal Processing:** Created the first-ever Graph Signal Processing framework (EDA-Graph) for electrodermal activity analysis, integrating spectral graph theory with deep learning for emotion recognition and affective computing (IEEE J. Biomed. Health Inform. 2024, IF: 6.7). Extended graph-based methodologies to electroretinogram signals for neurodevelopmental disorder classification, reducing computational complexity while maintaining diagnostic accuracy.
  - **Transformer & Attention Mechanisms for Sequential Data:** Developed transformer-based architectures for continuous emotion detection from multimodal physiological signals, implementing self-attention mechanisms for temporal feature extraction. Applied BERT and custom transformer models for clinical text mining and socioeconomic forecasting with ensemble methods achieving superior MAPE metrics in government analytics platforms.
  - **Multi-Omics & Computational Proteomics:** Pioneered semi-supervised learning approaches combining SVM with Particle Swarm Optimization for GO-Term prediction, later enhanced with GraphSAGE-based protein interaction networks to predict metabolic pathway activation with probabilistic graph reasoning.
  - **Production-Scale AI Deployment:** Deployed PyTorch-based fraud detection pipeline processing 50M+ daily transactions with <30ms latency on Neo4J graph databases, reducing fraud by 34.6%. Led cross-functional teams in Azure ML solution development, implementing NLP transformers for text preprocessing and ensemble deep learning models for multi-scenario forecasting with Power BI integration.
  - **Large-Scale Graph Analytics**: Implemented GNN-based anomaly detection systems on Hadoop/Spark clusters processing 1B+ daily telecom transactions, utilizing SCALA for distributed graph analysis to identify erratic transaction patterns and optimize database clearance operations.

SELECTED PEER-REVIEWED PUBLICATIONS [11 OF 24 PUBLICATIONS] & INVITED TALKS

- **Graph Signal Processing & AI for Biomedical Signals Publications**

11. "Graph-Based Analysis of Electroretinograms for Reducing Computational Complexity and Classifying Neurodevelopmental Disorders"
    L. R. Mercado-Diaz, Y. R. Veeranki, and H. F. Posada-Quintero
    BSN 2025 *(Top Conference in Body Sensor Networks)*, *****Topic: graph-based neurodevelopmental disorder detection [Paper]***
10. "Graph-Based Multi-Modal MRI Analysis with Probabilistic Attention for Stroke Lesion Detection"
    L. R. Mercado-Diaz, D. Aguiar, and H. F. Posada-Quintero
    Neurocomputing *(IF: 5.5, JCR Q1 in Computer Science), 2025,* *****Topic: graph neural networks for medical imaging [Paper]***
09. "Artificial Intelligence Approaches for the Detection of Normal Pressure Hydrocephalus: A Systematic Review"
    L. R. Mercado-Diaz, N. Prakash, G. X. Gong, and H. F. Posada-Quintero
    Applied Sciences *(IF: 2.5, JCR Q2 in Engineering), 2025,* *****Topic: AI for neurological disorder detection [Paper]***
08. "Fractal Analysis of Electrodermal Activity for Emotion Recognition: A Novel Approach Using Detrended Fluctuation Analysis and Wavelet Entropy"
    L. R. Mercado-Diaz, Y. R. Veeranki, E. W. Large, and H. F. Posada-Quintero
    Sensors *(IF: 3.4, JCR Q2 in Engineering), 2024,* *****Topic: nonlinear signal processing for affective computing [Paper]***
07. "EDA-Graph: Graph Signal Processing of Electrodermal Activity for Emotional States Detection"
    L. R. Mercado-Diaz, Y. R. Veeranki, F. Marmolejo-Ramos, and H. F. Posada-Quintero
    IEEE J. Biomed. Health Inform. *(IF: 6.7, JCR Q1 in Medical Informatics), 2024,* *****Topic: pioneering graph signal processing for biomedical signals [Paper]***
06. "Emotional States Detection Using Electrodermal Activity and Graph Signal Processing"

L. R. Mercado-Diaz, Y. R. Veeranki, F. Marmolejo-Ramos, and H. F. Posada-Quintero
BSN 2024 *(Top Conference in Body Sensor Networks),* *****Topic: graph-based emotion recognition [Paper]***

- **Multimodal Deep Learning & AI for Healthcare Publications**

05. "A Multimodal Deep Learning Exploration for Pain Intensity Classification"
J. O. Pinzon-Arenas, **L. R. Mercado-Diaz (co-first)**, B. E. Faremi, and H. F. Posada-Quintero
ICMI 2025 *(CORE: Ranked A in Human-Computer Interaction),* *****Topic: multimodal affective computing [Paper]***

04. "Detecting Autism Spectrum Disorder and Attention Deficit Hyperactivity Disorder Using Multimodal Time-Frequency Analysis with Machine Learning Using the Electroretinogram from Two Flash Strengths"
S. M. Manjur, **L. R. Mercado-Diaz (co-first)**, I. O. Lee, et al.
J. Autism Dev. Disord. *(IF: 6.9, JCR Q1 in Psychology), 2025,* *****Topic: machine learning for neurodevelopmental disorders [Paper]***

03. "Autoencoder Based Nonlinear Feature Extraction from EDA Signals for Emotion Recognition"
Y. R. Veeranki, **L. R. Mercado-Diaz (co-first)**, and H. F. Posada-Quintero
IEEE MeMeA 2024 *(Top Conference in Medical Measurements),* *****Topic: deep learning for emotion recognition [Paper]***

02. "Nonlinear Signal Processing Methods for Automatic Emotion Recognition Using Electrodermal Activity"
Y. R. Veeranki, **L. R. Mercado-Diaz (co-first)**, R. Swaminathan, and H. F. Posada-Quintero
IEEE Sens. J. *(IF: 4.3, JCR Q1 in Engineering), 2024,* *****Topic: nonlinear methods for affective computing [Paper]***

01. "Deep Learning Analysis of Electrophysiological Series for Continuous Emotional State Detection"
J. O. Pinzon-Arenas, **L. R. Mercado-Diaz (co-first)**, et al.
ACIIW 2023 *(Top Workshop in Affective Computing),* *****Topic: deep learning for continuous emotion detection [Paper]***

INDUSTRY EXPERIENCE (8-YEARS EXPERIENCE)

- **DATAKNOW – Data Science & Machine Learning Consultancy** — **Colombia**
Role: Senior Data Scientist & ML/AI Consultant — Dec. 2021–Aug. 2022
Led machine learning and R&D initiatives across multiple enterprise clients (Bancolombia, S.A.S., Scotiabank, Argos, Medtronic, Grupo Familia), delivering scalable AI solutions and ensuring best practices in model deployment and system integration.

Key Projects & Achievements:

- **Medtronic – Healthcare Knowledge Graph Architecture:** Designed and implemented Neo4J graph database consolidating clinical histories, medical equipment, and device data. Developed graph-based analytics for interaction analysis and market trend prediction in medical device ecosystem.
- **Argos – Deep Learning CI/CD Pipeline:** Architected end-to-end ML pipeline in Azure Databricks (Spark 3.0) for collections prediction and financial forecasting. Established continuous integration/deployment workflows using Azure DevOps for automated model retraining and versioning.
- **Puntos Colombia – Graph-Enhanced Recommendation System:** Built hybrid recommendation engine combining factorized matrix decomposition with SVM classifiers, optimized via gradient descent for probabilistic multi-category product scoring.
- **Scotiabank – Enterprise Data Warehouse Modernization:** Led complete data model transformation for pricing analytics, implementing ETL processes and mapping requirements across Azure Analytics platforms.
- **Family Group/Gluky – Cloud ML Migration:** Orchestrated on-premises to Azure Cloud migration, deploying microservices architecture for ML models on Databricks with DevOps automation.

- **TCS CENTRAL SOLUTION – Big Data & Cloud Analytics Division** — **Colombia**
Role: Big Data Analyst - Data Scientist & GNN Specialist — Jun. 2021–Dec. 2021
Specialized in large-scale graph analytics for telecom enterprise processing 1B+ daily transactions on Hadoop clusters using SCALA and Spark.

Key Achievements:

- **Graph Neural Network Anomaly Detection:** Architected and deployed GNN-based system for identifying erratic transaction patterns in real-time, enabling faster database clearance and eliminating visualization system errors
- **Multi-Source Data Integration:** Performed advanced analytics across heterogeneous data sources (relational and non-relational), implementing distributed graph processing algorithms for pattern recognition at scale
- **Predictive Maintenance Platform:** Contributed to enterprise-wide analytics platform combining traditional ML with graph-based relationship modeling for equipment failure prediction

- **BICODE – Cloud AI Solutions Consultant** — **Colombia**
Role: Azure Machine Learning Solution Specialist & Transformer Implementation Lead — Oct. 2021–Dec. 2021
Led 3-month government analytics project (DAPRE) developing end-to-end ML pipeline for socioeconomic indicators forecasting, managing data science team.

Key Achievements:

- **Transformer-Based NLP Pipeline:** Implemented BERT-based transformer models for topic extraction and semantic search across government datasets, enabling intelligent information retrieval from citizen feedback
- **Generative Text Preprocessing:** Deployed generative models for automated comment filtering and text normalization, improving downstream data quality for analytics teams

- **Ensemble Forecasting System:** Designed hybrid architecture combining traditional forecasting methods with neural networks (LSTM/Transformer), implementing automated model selection based on MAPE/MAE metrics with Power BI integration
- **Multi-Database ETL Architecture:** Built scalable data extraction pipelines interfacing with relational (SQL) and non-relational (MongoDB) databases for unified analytics

▪ **IUVITY-Aquila – Financial Technology & Fraud Analytics** **Colombia**
Role: Data Scientist – Graph ML & Real-Time Systems Engineer Apr. 2020–May. 2021
Developed production-grade graph-based fraud detection system processing 50M+ daily transactions, working within specialized Data Science team separate from BI operations.

Key Achievements:

- **Knowledge Graph Fraud Detection System:** Architected Neo4J graph database capturing multi-country fraud patterns and transaction relationships. Implemented PyTorch-based GNN pipeline achieving 34.6% fraud reduction with 30ms inference latency
- **Real-Time ML Pipeline Deployment:** Led end-to-end implementation from model development to production deployment, including system administration, performance optimization, and continuous monitoring
- **Big Data Infrastructure Leadership:** Directed Hadoop cluster implementation using Spark for historical transaction feature engineering across multiple banking institutions, enabling advanced fraud pattern analysis
- **Graph Visualization Analytics:** Developed interactive Knowledge Graph visualizations for fraud estimate tracking across international operations, supporting risk management decision-making
- **Database Architecture:** Designed and administered MongoDB for transactional data and Neo4J for relationship-based fraud analytics, implementing ETL pipelines for data integration

Technical Stack: PyTorch, Neo4J, MongoDB, Hadoop, Spark, Python, Graph Neural Networks

▪ **SBPM – Libraries and Archive and Digital Innovation** **Colombia**
Role: Digital Culture Articulator May. 2017–Apr. 2020
Served on Directive Committee leading Digital Culture Area, managing 102-person cross-functional team (92 Digital Culture Mediators, 4 Maker Space Leaders, 4 R&D Data Scientists, 1 BiblioLabs Project Leader) while directing innovation, R&D, data analysis, and educational initiatives.

Key Achievements:

- **Knowledge Graph for Library Systems:** Pioneered graph-based semantic linking architecture connecting books, topics, authors, and contextual information. Reduced literature discovery complexity for library staff by implementing intelligent recommendation pathways through Knowledge Graph traversal
- **R&D Innovation Leadership:** Directed research initiatives exploring graph database applications for cultural heritage preservation and information retrieval, bridging academic research with practical library operations
- **Maker Space Program Development:** Established and led hands-on technology education labs fostering digital literacy and innovation across community
- **Cross-Functional Team Management:** Coordinated teaching activities, research programs, and innovation projects across diverse teams, ensuring alignment between technological solutions and organizational mission

Technical Focus: Knowledge Graphs, Semantic Networks, Recommendation Systems, Graph Database Architecture

▪ **Instituto Tecnológico Metropolitano (ITM) – Biomedical Engineering Department** **Colombia**
Role: Teaching Assistant Jan. 2017–Dec. 2021
Instructed undergraduate-level courses in Digital Electronics II and Differential Equations, developing curriculum materials and supporting student learning in core engineering fundamentals.

▪ **Instituto Tecnológico Metropolitano (ITM) – Intelligent Machines and Pattern Recognition Laboratory** **Colombia**
Role: Research Undergraduate Student – Computational Proteomics & Graph ML Jan. 2017–Dec. 2021
Conducted foundational research in computational biology, developing novel graph-based approaches for protein function prediction and metabolic pathway analysis.

Key Research Contributions:

- **Graph-Based GO-Term Prediction:** Developed hybrid approach combining semi-supervised SVM (optimized via Particle Swarm Optimization) with GraphSAGE-based graph neural networks for Gene Ontology term prediction
- **Protein Interaction Network Analysis:** Implemented GNN architectures to model protein-protein interactions and predict metabolic pathway activation, computing probabilistic scores for interaction likelihood and functional annotation
- **Comparative Methodology Study:** Conducted systematic evaluation comparing traditional machine learning approaches with graph-based deep learning methods, demonstrating superior performance of GraphSAGE for capturing relational patterns in biological networks
- **Computational Bioinformatics Pipeline:** Built end-to-end pipeline for protein function prediction integrating sequence features, structural information, and network topology through graph representation learning

Technical Stack: Python, R, PyTorch, GraphSAGE, Semi-supervised Learning, Graph Neural Networks, Bioinformatics

SELECTED ACADEMIC PROJECT EXPERIENCE

- **Project: "Pioneering Graph Signal Processing Methods for Biomedical Signal Analysis and Neurological Disorder Detection"** — **Posada Q Lab**, UConn, CT, USA, Aug. 2022 – Current

  Role: Biosensing AI Research
- **Project: "Graph Neural Networks for Stroke Detection and Brain Structure Segmentation using Multimodal MRI"**

  Role: Biosensing AI Research
- **Project: "Multimodal Deep Learning for Real-World Cardiac Arrhythmia Detection using Smartwatch Photoplethysmography"**

  Role: Biosensing AI Research & Bayesian ML Researcher
- **Project: "Graph-Based Drug-Protein Interaction Prediction: ZIP11 Protein Binding Analysis"**

  Role: Biosensing AI Research & Graph Methods Researcher
- **Project: "Multimodal NLP/LLM Foundation AI for Drug Discovery and Protein Interaction Prediction"**

  Role: NLP/LLM & Biomedical AI Researcher
- **Project: "Foundational Multimodal Physiological Signal Processing: Integrating SKNA, ECG, and EDA for Comprehensive Health Monitoring"**

  Role: Biosensing AI Research
- **Project: "Foundational Multimodal Physiological Signal Processing: Integrating SKNA, ECG, and EDA for Comprehensive Health Monitoring"**

  Role: Biosensing AI Research
- **Project: "Foundational Multimodal Physiological Signal Processing: Integrating SKNA, ECG, and EDA for Comprehensive Health Monitoring"**

  Role: Biosensing AI Research
- **Project: "AI-Driven Gesture Recognition and Emotion Detection using Computer Vision"** — ESTIA, France, Jun. 2024 – Aug. 2024

  Role: Research Intern, Computer Vision & AI Researcher
- **Project: "Neuroimaging Analysis of Reading Comprehension: Structural Brain Correlates and Cognitive Training Using EEG-tDCS"** — **BIRC**, UConn, CT, USA, Aug. 2025 – Current

  Role: Research Software Engineer, iBrain Fellows Program

HUMAN-CENTERED DATA COLLECTION CONTRIBUTION (DESIGN/MANAGEMENT)

- **IRB Protocol Design and Management**
  - "Trust in Artificial Intelligence: An Empirical Study of Trust Dynamics in AI Interactions" (Approved in 2025). Designed experimental protocol and managed data collection
  - "OIME - Objective Integrated Multimodal Electrophysiological Index for the Quantification of Visceral Pain" (Approved in 2023). Designed comprehensive protocol integrating multiple physiological signals (EDA, ECG, facial expressions) for objective pain assessment.

RELEVANT QUALIFICATIONS & SKILLS

- Deep learning (DL), including large-scale training and fine-tuning of CNN, DNN, LSTM, Transformer, etc. (7+ years); Graph Signal Processing (7+ years), including ASR, NLP/LLM (6+ years), including **BERT/GPTs/Gemini**; wearable sensing (7+ years), including EDA, IMU, PPG, MRI, ERG; computational genomics (2 years). Computer vision, including localization, video gesture recognition, and face detection (2+ years).

SELECTED HONORS AND AWARDS [4 OF 9 HONORS]

- **Fellowship Recipient:** Research Software Engineer iBrain Fellows Program, CT, USA, 2025-2026
- **Doctoral Student Travel Award**, BSN 2025, CA, USA, Nov. 2024.
- **Fellowship Recipient:** TRANSCEND grant project recipient, CT, USA, 2024-2025
- **Doctoral Student Travel Award**, BSN 2024, IL, USA, Oct. 2024.
- **Student Travel Award**, ACII 2023, MA, USA, Sep. 2023.
- **Highest Honors in 2021 Graduation of Basic Applied Sciences,** Instituto Tecnologico Metropolitano, Colombia, Dec. 2022.